When θ is a vector of dimension $k>1$, the score function $u: \Theta \to \mathbb{R}^k$ with the i-th component

$$u_i(\theta; \underline{x}) = \frac{\partial}{\partial \theta_i} l(\theta; \underline{x}) \quad , \quad \theta \in \Theta$$

Note $\frac{\partial}{\partial \theta_i}$ means "differentiate with respect θ_i treating the θ_j, $j \neq i$ as constants"

The expected information is now a symmetric $k \times k$ matrix

$$I(\theta) = E_\theta \left[u(\theta; \underline{x}) \, u(\theta; \underline{x})^T \right]$$

This is the covariance matrix of the score function and is positive semidefinite (STAT2003 - pg 186)

The observed information matrix is

$$J(\theta) = \frac{-\partial^2}{\partial \underline{\theta} \partial \underline{\theta}^T} l(\theta)$$

meaning that

$$(J(\theta))_{ij} = \frac{-\partial^2}{\partial \theta_i \partial \theta_j} l(\theta) \qquad i, j = 1, \ldots, k.$$

As in the 1-d case

$$I(\theta) = E_\theta J(\theta) = -E_\theta \left(\frac{\partial^2}{\partial \theta \partial \theta^T} l(\theta; X) \right)$$

Consistency & asymptotic normality of $\hat{\theta}$ holds under regularity assumptions

$$\sqrt{n} \left(\hat{\theta} - \theta_0 \right) \rightsquigarrow N(0, i^{-1}(\theta))$$

where $i(\theta)$ is the expected information for a single observation.

$\theta = (\mu, \sigma^2)$

Example (Normal) Suppose $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$, $(\mu, \sigma^2) \in \mathbb{R} \times (0, \infty)$

$$l(\theta) = -\frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum_{j=1}^{n}(x_j - \mu)^2 - \frac{n}{2}\log(2\pi)$$

The score function is

$$\frac{\partial l}{\partial \mu}(\theta) = \frac{+1}{2\sigma^2}\sum_{j=1}^{n}(x_j - \mu)$$

$$\frac{\partial l}{\partial \sigma^2}(\theta) = \frac{-n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2}\sum_{j=1}^{n}(x_j - \mu)^2$$

The observed information is

$$\frac{\partial^2}{\partial \mu^2} l(\theta) = \frac{-n}{\sigma^2}, \quad \frac{\partial^2}{\partial \sigma^2 \partial \mu} l(\theta) = \frac{-1}{(\sigma^2)^2}\sum_{j=1}^{n}(x_j - \mu)$$

and $$\frac{\partial^2 l}{\partial (\sigma^2)^2}(\theta) = \frac{n}{2(\sigma^2)^2} - \frac{1}{(\sigma^2)^3}\sum_{j=1}^{n}(x_j - \mu)^2$$

~~The expected information~~

$$J(\theta) = \begin{bmatrix} n/\sigma^2 & \frac{1}{(\sigma^2)^2}\sum_{j=1}^{n}(x_j - \mu) \\ \frac{1}{(\sigma^2)^2}\sum_{j=1}^{n}(x_j - \mu) & \frac{-n}{2(\sigma^2)^2} + \frac{1}{(\sigma^2)^3}\sum_{j=1}^{n}(x_j - \mu)^2 \end{bmatrix}$$

(x_j over the sums)

The expected information

$$I(\theta) = E_\theta[J(\theta; X)] = \begin{bmatrix} \frac{n}{\sigma^2} & 0 \\ 0 & \frac{-n}{2(\sigma^2)^2} + \frac{1}{(\sigma^2)^3} \cdot n\sigma^2 = \frac{n}{2(\sigma^2)^2} \end{bmatrix}$$

what is $E[(X-\mu)^2] = \sigma^2$

Suppose $g : \Theta \to \mathbb{R}$ is 'smooth' transformation. Then

$$\sqrt{n}\,(g(\hat{\theta}) - g(\theta_0)) \xrightarrow{d} N(0,\ \nabla g(\theta_0)^T\, I^{-1}(\theta_0)\, \nabla g(\theta_0))$$

where $$\nabla g(\theta) = \begin{pmatrix} \frac{\partial g}{\partial \theta_1} \\ \vdots \\ \frac{\partial g}{\partial \theta_n} \end{pmatrix}$$

So for the $N(\mu, \sigma^2)$ model, suppose we want to estimate $\sigma = \sqrt{\sigma^2}$. The MLE is $\hat{\sigma} = \sqrt{\hat{\sigma}^2}$ and the asymptotic variance of $\hat{\sigma}$ is ...

$$g(\mu, \sigma^2) = \sqrt{\sigma^2}$$

$$\nabla g = \begin{pmatrix} \frac{\partial g}{\partial \mu} \\ \frac{\partial g}{\partial \sigma^2} \end{pmatrix} = \begin{pmatrix} 0 \\ \frac{1}{2}(\sigma^2)^{-1/2} \end{pmatrix} = \begin{pmatrix} 0 \\ \frac{1}{2\sigma} \end{pmatrix}$$

$$\nabla g^T\, i^{-1}\, \nabla g = \begin{bmatrix} 0 & \frac{1}{2\sigma} \end{bmatrix} \begin{bmatrix} \sigma^2 & 0 \\ 0 & 2\sigma^4 \end{bmatrix} \begin{bmatrix} 0 \\ \frac{1}{2\sigma} \end{bmatrix}$$

$$= \begin{bmatrix} 0 & \sigma^3 \end{bmatrix} \begin{bmatrix} 0 \\ \frac{1}{2\sigma} \end{bmatrix} = \frac{\sigma^2}{2}.$$

$$\sqrt{n}\,(\hat{\sigma} - \sigma) \xrightarrow{d} N(0, \sigma^2/2).$$